

Experiment No. 3

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Experiment Title:

Implement control structures and form validation; create a grading system based on user-entered marks.

Software / Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- CSS3
- JavaScript

Experiment Program Code

Files Used:

- grade.html
- script4.js
- style3.css
- **grade.html**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Grade Calculator</title>
  <link rel="stylesheet" href="style3.css">
</head>
<body>

<div class="container">
  <h1>Student Grade Calculator</h1>

  <input type="text" id="name" placeholder="Enter Student Name">

  <input type="number" id="english" placeholder="English Marks">

  <input type="number" id="maths" placeholder="Maths Marks">

  <input type="number" id="science" placeholder="Science Marks">

  <button onclick="calculateGrade()">Calculate Grade</button>

  <div id="result"></div>

</div>

<script src="script4.js"></script>

</body>
</html>
```

• script4.js

```
function calculateGrade() {

  let name = document.getElementById("name").value;

  let english = Number(document.getElementById("english").value);
  let maths = Number(document.getElementById("maths").value);
  let science = Number(document.getElementById("science").value);

  if(name==" " || english<0 || maths<0 || science<0 ||
  english>100 || maths>100 || science>100){

    document.getElementById("result").innerHTML =
    "<h3>Please enter valid details.</h3>";

    return;
  }

  let total = english + maths + science;
  let average = total / 3;

  let grade;

  if (average >= 50 && average < 66) {
```

```

    grade = "C";
}
else if (average >= 66 && average < 81) {
    grade = "B";
}
else if (average >= 81 && average <= 90) {
    grade = "A";
}
else if (average > 90) {
    grade = "A+";
}
else {
    grade = "Fail";
}

document.getElementById("result").innerHTML = `
    <h2>Result</h2>
    <p><strong>Student Name:</strong> ${name}</p>
    <p><strong>English:</strong> ${english}</p>
    <p><strong>Maths:</strong> ${maths}</p>
    <p><strong>Science:</strong> ${science}</p>
    <p><strong>Total Marks:</strong> ${total}/300</p>
    <p><strong>Average:</strong> ${average.toFixed(2)}</p>
    <p><strong>Grade:</strong> ${grade}</p>
`;
}

```

• style3.css

```

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
}

body{
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
    background:linear-gradient(135deg,#6a11cb,#2575fc);
}

.container{
    width:420px;
    background:ffffff;
    border-radius:20px;
    padding:30px;
    box-shadow:0 15px 35px rgba(0,0,0,0.25);
    animation:fadeIn 0.8s ease;
}

@keyframes fadeIn{
    from{
        opacity:0;
        transform:translateY(20px);
    }
    to{
        opacity:1;
        transform:translateY(0);
    }
}

h1{
    text-align:center;
    color:#4a148c;
    margin-bottom:25px;
    font-size:30px;
}

input{
    width:100%;
    padding:13px;
    margin:10px 0;
    border:2px solid #d9d9d9;
    border-radius:10px;
    outline:none;
    font-size:16px;
    transition:0.3s;
}

input:focus{
    border-color:#6a11cb;
    box-shadow:0 0 10px rgba(106,17,203,0.3);
}

button{
    width:100%;
    padding:14px;
    margin-top:15px;
    border:none;
    border-radius:10px;
    background:linear-gradient(90deg,#ff512f,#dd2476);
    color:white;
    font-size:18px;
    font-weight:bold;
}

```

```

        cursor:pointer;
        transition:.3s;
    }

    button:hover{
        transform:translateY(-3px);
        box-shadow:0 10px 20px rgba(221,36,118,0.35);
    }

    button:active{
        transform:scale(.98);
    }

    #result{
        margin-top:25px;
        background:#f8f5ff;
        border-left:6px solid #6a11cb;
        border-radius:12px;
        padding:18px;
        line-height:2;
        font-size:17px;
        color:#333;
    }

    #result h2{
        text-align:center;
        color:#6a11cb;
        margin-bottom:10px;
    }

    #result p{
        padding:6px 0;
        border-bottom:1px solid #e5e5e5;
    }

    #result p:last-child{
        border-bottom:none;
        font-size:22px;
        font-weight:bold;
        color:#ff5722;
    }

```

Application Description

The Student Grading System accepts the student's name and marks for English, Maths, and Science. JavaScript validates the entered data and uses control structures to calculate the total and average marks, determine the grade, and display whether the student has passed or failed. The generated result is shown dynamically below the form.

Features Implemented

Feature	Description
Student Input	Accepts student name and subject marks
Form Validation	Checks entered values before generating the result
Control Structures	Uses conditional statements to determine grade and result
Total Calculation	Calculates total marks out of 300
Average Calculation	Calculates the student's average percentage
Grade Evaluation	Assigns a grade according to the calculated average
Result Status	Displays PASS or FAIL based on entered marks
DOM Manipulation	Displays the generated result dynamically

Output

Rahul Narwade
PRN: 24070521046

Student Grading System

Rahul Narwade

Subject	Marks
English	87
Maths	67
Science	57

Generate Result

Name : Rahul Narwade
Total : 211/300
Average : 70.33%
Grade : B
Result : PASS

Output Summary

- Accepts marks for English, Maths, and Science.
- Calculates the total marks and average percentage.
- Determines the student's grade using control structures.
- Displays the final PASS/FAIL result.
- In the shown output, the total is 211/300, the average is 70.33%, the grade is B, and the result is PASS.

Result / Conclusion

The Student Grading System was implemented successfully using HTML, CSS, and JavaScript. The experiment demonstrated form validation and JavaScript control structures for processing user-entered marks. It successfully calculated total and average marks, assigned a grade, determined the pass/fail status, and displayed the result dynamically.